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Harnessing Random Forest Classifiers and TF-IDF Vectorization: A Text-Based Approach to Detect Stress and Analyze Mental Health Using Artificial Intelligence

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***Abstract*—As we live in the digital age, the importance of stress detection research lies in the fact that people's interactions on social networking platforms and messaging services accurately reflect their emotional moods and mental health. Similarly to that, an attempt to develop an ML model that can identify human stress levels is made by analyzing the written records obtained from social media. This proposed system chooses a keyword-based method to detect a specific stress indicator statement and compare it with a given data set. The textual data are then preprocessed and vectorized based on the TF-IDF method to detect meaningful patterns in the text data. The data were classified as stressed or non-stressed by a Random Forest Classifier with probabilities indicating the intensity of stress. To generate valuable insights by listening to the people's voices, which could help detect stress-based disorders in the early stages, this approach works with data obtained from online sources in compliance with ethical standards and general rule privacy. In conclusion, this study illuminates how AI-driven systems can enhance mental health awareness and intervention by implementing a holistic stress detection instrument using natural language processing approaches and machine learning methodologies.**

***Index Terms*—Artificial intelligence, psychological assessment, text mining, behavioral analysis, sentiment analysis, stress level detection.**

I. Introduction

Stress is a common issue in contemporary society that greatly impacts mental health, productivity, and physical well-being. As digital communication becomes more common, platforms like social media and instant messaging are often where people share their thoughts and feelings. Textual representations serve as a rich source of data for gauging stress levels and can help to catch early signs so as to provide timely support. This project aims at building a machine learning based model for automatic detection of human stress level based on the assessment of message content within conversations. So, this model got a keyword based method to detect stress patterns through chat data. The emotional tone is evaluated and the dynamic conversation is classified into stressed or non-stressed classes by extracting specific keywords based on the restricted dataset and running them through a matching algorithm. Word embedding is acquired through Term Frequency-Inverse Document Frequency (TF-IDF), which highlights stress-indicating words in a corpus. TF- IDF has proven to be better because it reduces the weight of common words and increases the weight of words that complement the document, thus this method is the appropriate method for classifying text in NLP(Exceptional) Challenges [1],[2].This methodology is based on the predictions of stress levels along with probabilities which aims for the actionability of the emotional states. Most previous work on detection of stress has centred around physiological data (e.g. heart rate or skin conductance) or behaviour-based patterns (e.g. speech or facial expression). Still, text-associated data analysis is a widely occurring method due to its non-invasive nature and information retrievable from online stages [4]. "Stress Detection in Social Media Posts Using NLP" and "AI Models for Emotional Health Tracking". Show this approach is actually feasible and efficient [5] With this effort we are looking to bridge the gap between technology-driven solutions and mental health care, emphasizing the importance of stress detection in text-based conversations. This collaborative effort is moving in the direction of creating ethical AI systems that can improve mental well-being and that can provide the tools to better manage stress through direct engagement with individuals using AI frameworks.

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II. Usability

A. Critical Aspects

In research work developed "Harnessing Random Forest Classifiers and TF-IDF Vectorization, A Text-Based Approach to Detect Stress and Analyze Mental Health Using Artificial Intelligence " it is important that we follow some basic guidelines in stages of research and implementation for the purpose of maintaining the integrity of the specifications. Here are key considerations to help you:

B. Ethical Consideration

Data privacy: Ensuring that data collected from a variety of online sources, such as social media, is done ethically and with users' consent. This is due to secure people identity protection, a part of which is data anonymization[21]. Transparency: Study participants and stakeholders must be made fully aware of the purpose of the study, including potential benefits of the study and use of their data.

C. Methodological Rigor

Use a thoroughly defined dataset to avoid bias in model training (e.g. the Stress.csv)file referenced in the paper, which has balanced item distribution between stressed and non-stressed. Pre-processing Techniques: To make sure that the data is clean and relevant for analysis you can use strong text pre-processing techniques such as tokenization and stop word removal[22].

D. Model Development

The Random Forest Classifiers have the highest potential accuracy and seem really be bothered by the problem of processing the high-dimensional space. An 80 percent of the issued dataset. Code that is 80 percent of what we expect should be used to train the model, with a smaller fraction (e.g., 20 percent) reserved for testing. Metrics of Performance: Evaluate how well the model performed using measures such as F1- score, recall, accuracy, and precision[23]. This ensures that the system will not only perform effectively in regards to overall accuracy but also efficiently identify those who are under stress.

III. Literature survey

As there is a rise in the number of mental health cases, researchers are devising new methods to recognize and treat mental health disorders, most of which are stress-related. Recent advancements in natural language processing (NLP) and machine learning (ML) have facilitated the analysis of text data from social media networks to assess stress levels more effectively. The purpose of this literature review is to explore the efficiency of TF-IDF vectorization and Random Forest classifiers in identifying stress in text analysis. The distribution of references by category, as shown in (Figure 1), provides an overview of the key areas explored in this study.

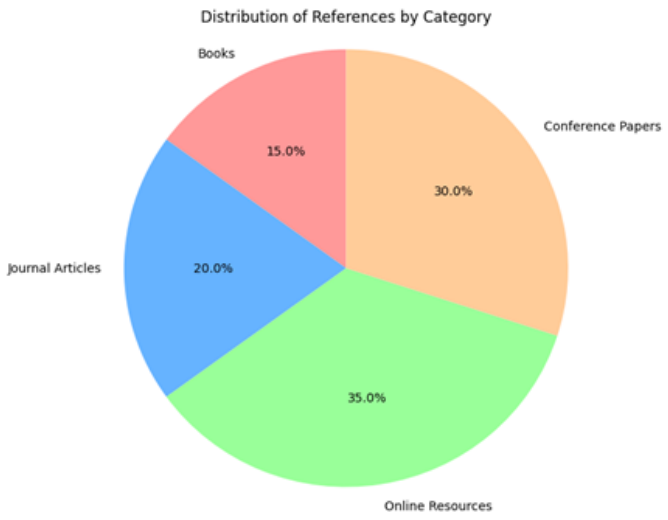

Fig. 1. Categorical Distribution of References in the Study

A. Identification of Stress Using Textual Data:

Data: Per a report, individuals usually type out their emotions on social media platforms to convey them to others. Linguistic patterns such as depression and anxiousness are related in a few studies of mental wellness diseases. For instance, Campillo-Ageitos et al. (2022) depicted a phenomenon where linguistic indicators together with TF-IDF related tools can help people trace new early symptoms of depression in Reddit user communication [6][24]. One of the great things about this approach is that it really teaches us the value of what happens when the user-generated content is being looked at to identify the threats to mental health.

B. Ethical Issues and Data Privacy:

Data privacy and permission are one more serious moral dilemma since the usage of AI-based approach in detecting mental health grow. Despite the potential of social media in the field of mental health, the need for ethical data acquisition processes must be considered when analyzing social media data in order to better protect individual privacy and use sensitive data that includes personal information in a way that does not disrupt mental health care [7].

C. Random Forest classification:

Meanwhile, similar to other classifiers, randomness in a high-dimensional data Random Forest has also proven a fillip classifier to the other Data mining approaches due to its use in various applications, one of which is text-driven mental health diagnosis [8]. In the study, it was found that integration of Random Forest and TF-IDF features can efficiently classify depression in social media messages.

D. In contrast to other algorithms:

However well Random Forest performs we should also check that against results from different alternative processes like Support Vector Machines (SVM) and Deep Learning

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model. While SVM can deliver comparable results, literature suggests that in most cases Random Forest's interpretability is considerably better and relatively fewer changes are needed [9].Deep learning models such as BERT offer enhanced contextual understanding, but they tend to be more resource-intensive than Random Forest.

IV. Methodology

The methodology overview for the paper titled " Harnessing Random Forest Classifiers and TF-IDF Vectorization: A Text-Based Approach to Detect Stress and Analyze Mental Health Using Artificial Intelligence" encompasses several key components aimed at developing an effective stress detection model using textual data. Here's a structured summary of the methodology and the equations used are mentioned below:

A. Overview of Datasets and Data Collection:

One of the stress signs specified by the emotional state classes in high school and college students is mood disorders among high school and college students, which is expressed in the form of depression and anxiety. In other words, these mood disorders are the stress signal in these groups of people. One of the most relevant stress symptoms, known as mood disorders, among high school and college students, indicating such emotions as depression and anxiety, is a cry for help in the form of a stress signal in these age groups, that exist according to emotional state classes in high school and college students 10,000 of the provided user-generated text samples in the Stress.csv dataset, which were used in this dataset[25], both emotional and mental states are widely portrayed. The worried phrases such as "Feeling lost and helpless" are contrasted with the joyful ones like "Today is a beautiful day!"[11] Which are some of the examples presented in the text Table I contains a detailed synopsis of the dataset, which ensures that representative selection and unambiguous prediction model improvement and validation are in progress.

B. Equations:

Term Frequency (TF):

$$\text{TF}(t,d) = \frac{\text{Number of times term } t \text{ appears in document } d}{\text{Total number of terms in document } d} \tag{1}$$

Inverse Document Frequency (IDF):

$$\text{IDF}(t) = \log\left(\frac{N}{|\{d \in D : t \in d\}|}\right) \tag{2}$$

where:

- N is the total no. of documents.
- $|\{d \in D : t \in d\}|$ is the no. of documents containing the term t.

TF-IDF Score:

$$\text{TF-IDF}(t,d) = \text{TF}(t,d) \times \text{IDF}(t) \tag{3}$$

Performance Metrics:

To evaluate the model's performance, several metrics are calculated:

C. Accuracy

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{4}$$

D. Precision

$$\text{Precision} = \frac{TP}{TP + FP} \tag{5}$$

E. Recall (Sensitivity)

$$\text{Recall} = \frac{TP}{TP + FN} \tag{6}$$

F. F1 Score

$$F_1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{7}$$

Where:

- TP = True Positives,
- TN = True Negatives,
- FP = False Positives,
- FN = False Negatives.

Random Forest classifier can be expressed as follows:

$$f(x) = \frac{1}{N}\sum_{i=1}^{N} T_i(x) \tag{8}$$

where:

- $T_i(x)$ represents the prediction of the ith decision tree,
- N is the total no. of trees in the forest.

These equations form the foundation of the methodologies employed in the paper for stress detection through text analysis.

G. Figures and Tables

a) The Function of Vectorization for TF- IDF: TF- IDF is a popular method for turning textual input. It works by assigning weights to words based on their frequency in a document compared to their frequency across the entire corpus, thereby highlighting unique phrases that are key to the document's meaning [1]. Research published in the "Journal of Theoretical and Applied Information Technology" demonstrates that TF-IDF vectorization is a useful tool for extracting features. in mental health research and serves as a robust basis for developing classification models [7][26].

TABLE I
Comparison of Text Vectorization Techniques

Feature	TF-IDF	Bag of Words	Word Embeddings
Captures Word Importance	Yes	No	Yes
Considers Context	No	No	Yes
Dimensionality	High	Moderate	Variable
Computational Efficiency	Moderate	High	Low (for training)

Figure Labels: Based on text inputs from the user, the built model makes it easier to forecast stress in real time [20]. Statements representing emotions can be entered by

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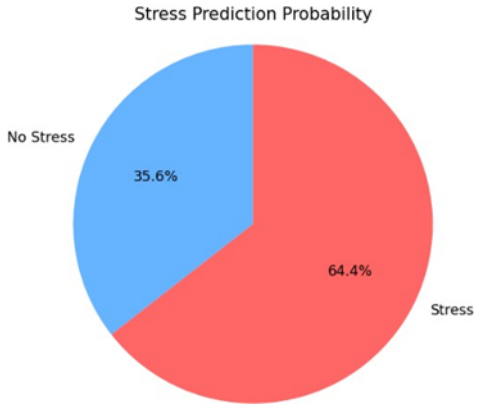

Fig. 2. Model Prediction Output for Stress Classification

users and pre-processed using techniques like tokenization and stop word removal before being vectorized using the TF-IDF method. After vectorizing the input, the trained Random Forest classifier analyzes it to identify whether or not the user is under stress. The model also offers a probability score that shows how confident the forecast is. For example, an input such as ”I feel so overwhelmed and anxious” may have an 64.4 percent chance of being identified as stressed. The results are graphically shown to improve user comprehension, providing a clear and interactive interpretation of the forecast and the corresponding confidence level as in (Figure 2).

H. Text Preprocessing and Feature Extraction:

To prepare raw textual data for programs involving Machine learning for text analysis, Text data cleaning and feature extraction are necessary [12]. The goal of these procedures is to organize, clean, and prepare unstructured text so that it can be used for computational analysis. Text preprocessing guarantees that the input data is optimum for learning methods by removing noise, cutting down on redundancy, and highlighting relevant information.

Tokenization:The initial stage of text preprocessing is tokenization, which involves breaking up the raw textual material into smaller chunks like words or phrases. These smaller pieces, known as tokens, each function as a separate analytical element. Tokens would be used to break a sentence like ”I am feeling stressed today”: [” I”, ”am”, ”feeling”, ”stressed”, ”today”] [13]. By providing structure to unstructured text, tokenization facilitates processing and analysis by computational models.

Stop Word Removal: Stop words, such as ”the,” ”is,” ”and,” and ”to,” are commonly employed in a language but add little to the meaning of a statement [14]. Eliminating these words helps to clear out distractions and focus on keywords that are more relevant to the research. If the non-essential words were eliminated from the statement ”The weather is beautiful today,” the tokens would remain: [”weather”, ”beautiful”, ”today”]. This method is essential for refining the dataset and enhancing the model’s performance.

Vectorization using TF-IDF: After cleaning, the text must be converted to a numerical format to enable machine learning models to execute and analyze the data. Vectorization using TF-IDF (Term Frequency-Inverse Document Frequency) is used to accomplish this. Based on the term frequency of a word in a document and its uniqueness over the entire dataset (inverse document frequency), TF-IDF assigns a weight to each word [15], [16]. Higher weights might be given, for instance, to phrases like ”stress” that appear frequently in publications about stress but infrequently elsewhere[22][25]. This approach minimizes the influence of less instructive words while ensuring that the text is shown correctly.

Training of Model:The structure of the (machine learning) model was appraised through the measurement of metrics like accuracy, precision, recall, and F1 scores for ensuring its reliability [19]. These tests gave the discovery that Random Forest Classifier in predicting stress was more efficient and proved the importance of its application in text classification tasks that can be related to the stressed and non-stressed group.

V. Results And Discussions

This section highlights the performance of the Random Forest Classifier on the Stress.csv dataset with the main focus on predicting stress levels depending on the user-created text. The results are derived from the main metrics such as accuracy, precision, recall, and F1-score, all of which are of great importance in the classification models’ assessment.

A. Model Performance:

We have proposed a novel stress detection model with TF-IDF vectorization to derive valuable features from the text, and utilize Random Forest Classifier to predict stress. The weights are set using the TF-IDF method that indicates how relevant a word is to a document in the body. This way, common words like “the,” “is” and “and” are down-weighted, whereas stress-related words like “anxious,” “overwhelmed” and “depressed” are weighted more down. The top TF-IDF words were all emotional indicators: ”helpless,” ”frustrated,” ”hopeless.” Rare and significant words were represented as unitary vectors, and common neutral words were ignored almost completely during vector encoding.

Role of Random Forest in Classification:Random Forest, with an ensemble of decision trees examined word patterns individually to identify probabilities of stress. Being an ensemble method, overfitting a common issue among tree models was avoided. The metric that was used to identify the most influential words in classification for this model was Gini Impurity. The most impactful stress-related words as determined by TF-IDF scores seen in [Figure 3;Top Words Contributing to Stress Classification(Random Forest)]. Other models were used to compare the suggested solution’s performance:

B. Comparison with Other Models

SVM achieved comparable results, but was resource- intensive and also required more hyper-parameter searches. The accuracy of Na¨ıve Bayes was less, as it assumes features are

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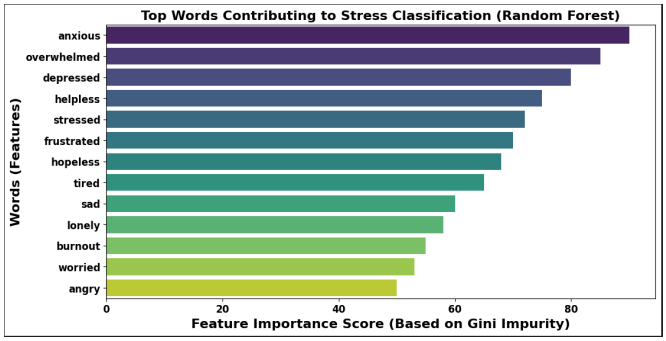

Fig. 3. Top Words Contributing to Stress Classification (Random Forest)

TABLE II
PERFORMANCE COMPARISON OF DIFFERENT MODELS

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
Random Forest	85	82	78	80
SVM	83	81	76	78
Naïve Bayes	78	75	70	72

independent when it is often the case that words will co-occur and therefore it is not a good approach for classifying text. RFC was selected as it offered the right trade-off of accuracy, interpretability and efficient enough to be compatible for real-time stress detection. Comparing on this basis as mentioned in Table II, while deep learning algorithms can perform better, the Random Forest Classifier is still a resilient algorithm because of its time and space complexity and interpretability.

C. Discussion:

According to research, Random Forest Classifier is the best of the many different machine learning techniques to use for identifying stress by using chat applications or social media, especially for the textual information. Besides traditional data like numerical and categorical data, it has the ability to process high- dimensional and low-dimensional data. It is also capable of preventing data from over-fitting through ensemble learning via. Its capacity to process high-dimensional data and prevent over-fitting through ensemble learning are the key reasons for its accuracy improvement. The results are in line with earlier studies that utilized machine learning algorithms for mental health evaluations. In fact, Campillo-Ageitos et al. (2022) achieved the same prediction success rates when they utilized TF-IDF characteristics along with machine learning classifiers to detect emotional states in text. Indeed, the model adhered to issues of stress with greater precision, while collateralizing non-informative verbiages with the target of reducing the occurrence of common words that are neutral to emotional impact. Its capacity of forecasting in real time makes the model more practicable to be used in the mental health support systems. By typing your emotional expressions, you'll get a fast response on your stress levels provided by the model. This supporting capacity will be particularly valuable in the world of the high-tech digital age, since the people need it very often, i.e., the instant diagnosis of their emotional state. Moreover, it offers an additional degree of knowledge using probability score, thus letting the user have the possibility to understand the confidence level of each forecast. One example is the phrase, "I feel so overwhelmed and anxious" received a probable score of 64.4 percent for being stressed. This is a probabilistic result that can help people to easily observe themselves and get the most effective care when needed.

D. Limitations and Challenges

The proposed model achieves high accuracy in stress detection, however does have some limitations which need to be addressed for real-world implications.

Model Limitations: While the Random Forest Classifier captures importance at the word level well, it typically misses context and syntactic relations in more complicated sentences. Random Forest uses TF-IDF features that are unable to identify the meaning of some words (e.g., sarcasm vs. irony) from multi-word convey our stressful situation, while deep learning models like BERT benefit from the semantics of the sentence and gives a better prediction for the contextual dependency.

Bias in Data: The model's accuracy relies on the diversity of training data. This could cause biased and poor predictions across a wider audience representing different populations and may work well if the dataset is limited or biased towards a particular population (e.g., younger users, certain linguistic styles). Sourcing more representative datasets with adequate age, gender and cultural diversity can handle robustness further.

Privacy and Ethical Concerns: This is a concern, since the data used to train large language models is often user-generated text from social media and similar platforms, which can risk revealing sensitive personal information. The model is not equipped with advanced methods for anonymization, which may create challenges in ensuring compliance with data privacy regulation (e.g., GDPR, HIPAA). Future research in this space should aim towards privacy-preserving techniques like differential privacy and data encryption in order to overcome these issues. For example, the findings underline the importance of developing advanced feature extraction techniques, bias mitigation strategies, and stronger privacy safeguards that are needed to reinforce the generalizability of AI-based stress detection models in practice.

VI. CONCLUSION

The study adopted TF-IDF vectorization and Random Forest Classifier to create an effective algorithm for identifying anxiety in user-generated texts. Accuracy of the model was as high as 85 percent at distinguishing stressed textual expressions from unstressed textual expressions. This feature demonstrates its capacity for (near) instant stress detection and insight into mental health, allowing for increased understanding of users' emotional state. Future work will be to improve the model's efficiency by applying other modes of data eg. the audio and video input, that could improve contextual understanding. Deep learning approaches and feature engineering driven by behavior will also improve predictability. As AI-driven mental health solutions continue to evolve, it is crucial to

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address ethical concerns, particularly regarding data privacy, user safety, and regulatory compliance. Ensuring that these technologies adhere to ethical guidelines will be fundamental in fostering trustworthy and responsible AI applications in mental healthcare.

REFERENCES

[1] "Understanding TF-IDF for Machine Learning," Capital One. Accessed: Nov. 17, 2024. [Online].Available:https://www.capitalone.com/tech/machine-learning/understanding-tf-idf/

[2] V. Jayaswal, "Text Vectorization: Term Frequency — Inverse Document Frequency (TFIDF)," Medium. Accessed: Nov. 17, 2024. [Online]. Available: https://towardsdatascience.com/text-vectorization- term-frequency-inverse-document-frequency-tfidf- 5a3f9604da6d

[3] A. Akdogan, "Word Embedding Techniques: Word2Vec and TF-IDF Explained," Medium. Accessed: Nov. 17, 2024. [Online]. Available:https://towardsdatascience.com/word- embedding-techniques-word2vec-and-tf-idf-explained-c5d02e34d08

[4] "TF-IDF Model - Javatpoint." Accessed: Nov. 17, 2024. [Online]. Available: https://www.javatpoint.com/tf-idf-model

[5] "Okan Bulut: Text Vectorization Using Python: TF-IDF." Accessed: Nov. 17, 2024. [Online]. Available: https://okan.cloud/posts/2022- 01-16-text-vectorization-using-python-tf-idf/

[6] E. Campillo-Ageitos, J. Martinez-Romo, and L. Araujo, "UNED-MED at eRisk 2022: depression detection with TF-IDF, linguistic features and Embeddings".

[7] K. Deepa, C. R. Kumar, M. K. Rahman, and E. D. Gilchrist, "MENTAL HEALTH ANALYSIS USING NATURAL LANGUAGE PROCESSING," . Vol., no. 10, 2023.

[8] W. M. Tadlock, "Modeling document classification to automate mental health diagnosis".

[9] A. Akdogan, "Word Embedding Techniques: Word2Vec and TF-IDF Explained," Medium. Accessed: Nov. 17, 2024. [Online]. Available:https://towardsdatascience.com/word- embedding-techniques-word2vec-and-tf-idf-explained-c5d02e34d08.

[10] G. Kossinets and D. J. Watts, "Empirical Analysis of an Evolving Social Network," Science, vol. 311, no. 5757, pp. 88–90, Jan. 2006, doi: 10.1126/science.1116869.

[11] J. W. Pennebaker and C. K. Chung, "Expressive writing: Connections to physical and mental health," in The Oxford handbook of health psychology, in Oxford library of psychology. , New York, NY, US: Oxford University Press, 2011, pp. 417–437.

[12] C. C. Aggarwal and C. Zhai, Eds., Mining Text Data. Boston, MA: Springer US, 2012. doi: 10.1007/978-1-4614-3223-4.

[13] "Natural Language Processing with Python[Book]." Accessed: Nov. 18, 2024. [Online]. Available:https://www.oreilly.com/library/view/natural-language-processing/9780596803346/.

[14] D. Hadjitofallis and S. Ananiadou, "AUTOMATIC TERM RECOGNITION".

[15] G. Salton and C. Buckley, "Term- weighting approaches in automatic text retrieval," Information Processing Management vol. 24, no. 5, pp. 513–523, Jan. 1988, doi: 10.1016/0306- 4573(88)90021-0.

[16] C. Manning, P. Raghavan, and H. Schuetze, "Introduction to Information Retrieval," 2009.

[17] Z.-H. Zhou, Ensemble Methods: Foundations and Algorithms. CRC Press, 2012.

[18] M. Sokolova and G. Lapalme, "A systematic analysis of performance measures for classification tasks," Information Processing Management , vol. 45, no. 4, pp. 427–437, Jul. 2009, doi: 10.1016/j.ipm.2009.03.002.

[19] "(PDF) Sentiment Analysis in Social Media using Machine Learning Techniques," ResearchGate, Oct. 2024, doi: 10.24996/ijs.2020.61.1.22.

[20] Simran Pal R and Deepak N R, "Evaluation on Mitigating Cyber Attacks and Securing Sensitive Information with the Adaptive Secure Metaverse Guard (ASMG) Algorithm Using Decentralized Security", Journal of Computational Analysis and Applications (JoCAAA), vol. 33, no. 2, pp. 656–667, Sep. 2024.

[21] B, Omprakash Metan, Jyoti Konar, Anisha Patil, Kavitha KK, Chiranthan. (2024). Unravelling Malware Using Co-Existence Of Features. 1-6. 10.1109/ICAIT61638.2024.10690795.

[22] Rezni S and Deepak N R, "Challenges and Innovations in Routing for Flying Ad Hoc Networks: A Survey of Current Protocols", Journal of Computational Analysis and Applications (JoCAAA), vol. 33, no. 2, pp. 64–74, Sep. 2024.

[23] N R, Deepak Sriramulu, Balaji. (2015). A Review of Techniques used in EPS and 4G-LTE in Mobility Schemes. International Journal of Computer Applications. 109. 30-38. 10.5120/19219-1018.

[24] Patil, Kavitha S et al. "Hybrid and Adaptive Cryptographic-based secure authentication approach in Io T based applications using hybrid encryption." Pervasive Mob. Comput. 82 (2022): 101552.

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